



Boeing Co (NYSE: BA)
April 2024

Investment Team
Industrials & Consumers

Research Team
Industrials & Consumers



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Company Overview

Company Highlights

- **History:** Boeing was established in 1916 by William Boeing and George Westervelt. Some key milestones in the company's history include the first flight of the B-17 in 1935, the introduction of the 747 in 1970, and the launch of the Dreamliner 787 in 2009.
- **Valuation:** Boeing's stock performance has fluctuated due to industry trends and company-specific events. The market factors in order backlogs, new aircraft launches, and defense contracts. Key investment highlights are Boeing's diverse portfolio, innovation, and global client relationships.
- **Strategy:** Boeing is focusing on safety, innovation, and efficiency. The company's strategy emphasizes new technologies, operational excellence, and expanding its global presence, with strong management commitment to delivering high-quality products.
- **Corporate Finance Transactions:** Boeing's recent notable acquisitions include Aurora Flight Sciences and KLX Aerospace Solutions.

Industry Data	Average	Revenue Mix	%	Gross (\$mm)
Commercial Airplanes				
EV/EBITDA -122.167	55.837	Commercial Airplanes	43.5%	\$33,901.0
EV/Revenue - 3.967	2.392	Defense, Space & Sec..	32.0%	\$24,933.0
P/E - 22.163	29.004	Gbl. Service & Support	24.5%	\$19,127.0
Defense, Space & Security				
EV/EBITDA - 92.594	-482.399	Gross Margin	9.93	
EV/Revenue - 5.527	9.831	ROIC	-4.50	
P/E -12.506	994.763	Debt to Equity	-2.818	
		Cash ratio	0.17	
		ROA	-1.62	
		Pretax Margin	-2.58	

Valuation & Share Performance

Key Valuation Statistics

Enterprise Value **\$143bn** Revenue (FY23) **\$77.8bn**

Market Cap	\$105bn	EBITDA (FY23)	\$1.1bn
P/E	N/A	EBITDA Margin '23	1.40%
EV/EBITDA	46.50x	Debt / EV	0.34
Cash	\$7.5bn	Total Debt	\$47.9bn

Share Price | Current: **\$174.00** | High/Low: **\$168.65 / \$164.93**





Boeing Business Model

Operating Segments

- Boeing generates revenues by designing, developing, manufacturing commercial jetliners, military aircrafts, satellites, missile defense, human space flight and launch systems and services.
- It has mainly 3 operating segments:
 - Commercial Airplanes: Makes and sells commercial jet aircraft and provides fleet support services, principally to the commercial airline industry and to individuals.
 - Defense, Space & Security Systems: Engages R&D, production and modification of manned and unmanned military aircraft and weapons systems.
 - Global Services: This segment provides services to Boeing's commercial and defense customers worldwide.

Key Figures as of May 2023

KEY INFORMATION ABOUT



Company
Boeing



Market Capital
\$120.86 billion



Public or private
Public



Headquarter
Arlington, Virginia



Profit
\$3.53B



CEO
Dave Calhoun



Foundation
Employees Community Fund (ECF)

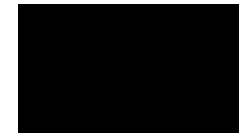


Revenue
\$62.2B



Number of Employees
140,000

Competitors

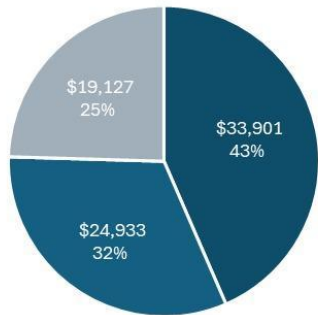




Operating Forecast

Revenue Drivers

Boeing Revenue Split 2023



■ Commercial Airplanes ■ Defense ■ Global Services

Revenue & EBITDA Forecast

(in Millions of Dollars)	2021	2022	2023	2024E	2025E
Commercial Airplanes	\$ 19,714	\$ 26,026	\$ 33,901	\$ 27,918	\$ 39,085
Defense, Space & Security	\$ 26,540	\$ 23,162	\$ 24,933	\$ 30,169	\$ 36,203
Global Services	\$ 16,328	\$ 17,611	\$ 19,127	\$ 22,952	\$ 27,543
Total Revenues	\$ 62,582	\$ 66,799	\$ 77,961	\$ 81,039	\$ 96,383
Commercial Airplanes Revenue Growth %		32.02%	30.26%	-17.65%	40.00%
Defense, Space & Security Revenue Growth %		-12.73%	7.65%	21.00%	20.00%
Global Services Revenue Growth %		7.86%	8.61%	20.00%	20.00%
Total Revenues Growth %		6.74%	16.71%	3.95%	18.90%
Total Adjusted EBITDA	-\$ 2,902	-\$ 3,519	-\$ 773	\$ 2,512	\$ 35.00
EBITDA Margin %	-4.64%	-5.27%	-0.99%	3.10%	0.04%

Commercial Airplanes: First Quarter revenue of \$4.7 billion and operating margin of (24.6) percent primarily reflect lower 737 deliveries and 737-9 grounding customer considerations

Even after the FAA audit on the 737, it managed to secure 125 net orders in 2024 Q1.

There exists a Backlog of \$448 Billion, suggesting a positive free cash flow for 2024

Defense and Space: Awarded 17 P-8 aircraft for Canada and Germany, Secured final contract from the U.S. Navy for 17 F/A-18 production aircraft
Q1 2024 was another quarter of sequential improvement. Backlog grew to \$61 Billion

Management plans to increase margins to high single digits by 2025 assuming a forecasted growth rate of 20%

Global Services: Q1 2024 was another strong quarter for this segment with a strong operating margin of 18.2%

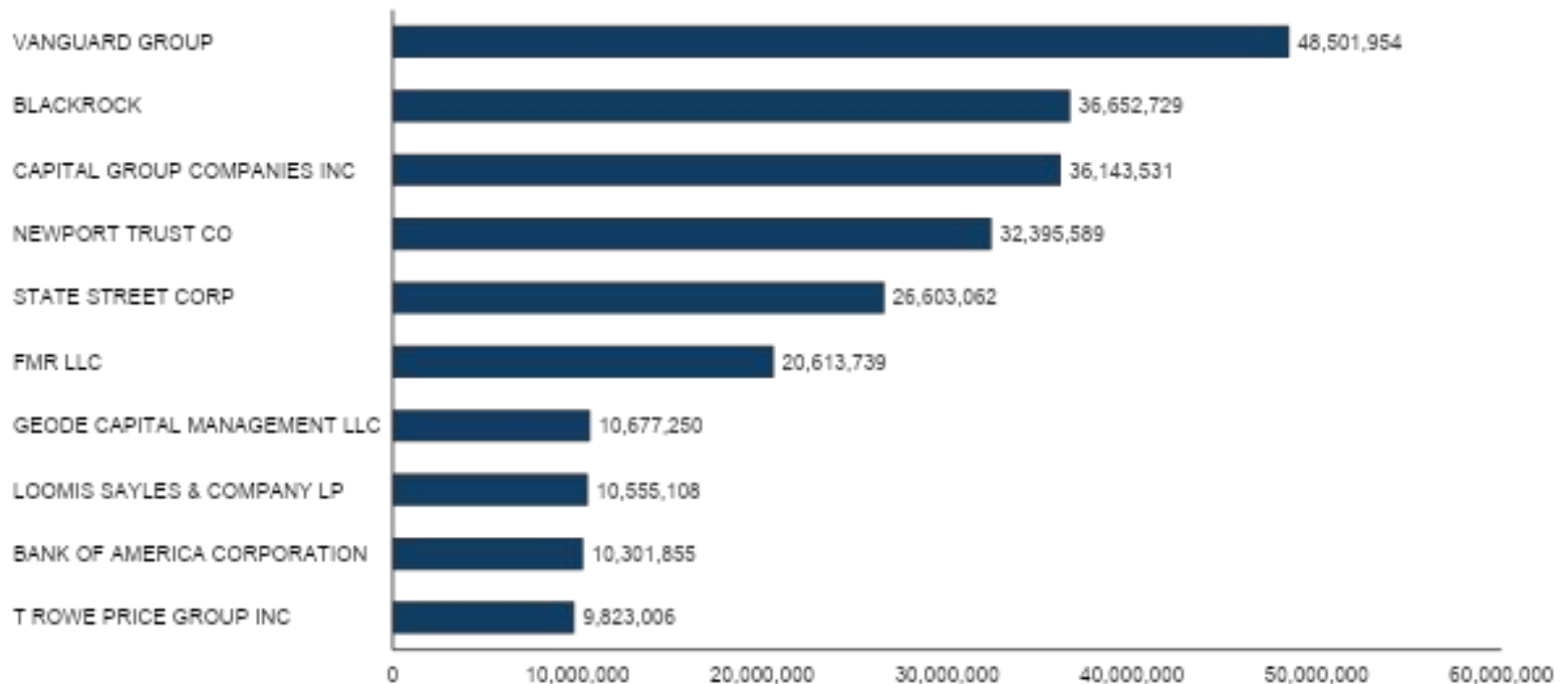
Boeing opened another maintenance facility in Jacksonville, Florida, to support military customers



Shareholder Ownership Analysis

Ownership Analysis

Boeing's ownership structure reveals a significant concentration of institutional investors, with major players like Vanguard Group and BlackRock leading the pack. The high percentage of institutional shareholders indicates stability, with a long-term holding pattern, suggesting that float turnover is relatively low. Insiders, including executives and board members, also hold shares, but to a lesser extent. Retail investors account for a smaller share.





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Competitive Landscape

Company	EV/ LTM Sales	EV/ LTM EBITDA	Strategy	Structure
 AIRBUS	EV/Sales: 1,81 P/E: 28,54	EV/Revenue: 1.72 EV/EBITDA: 14.33	Airbus recently increased production rates for A320 family to meet high demand. They also focus on developing hydrogen-powered aircraft. Analysts see strong commercial business but are cautious on the defense sector. Their growth strategy consist around the exploration of new technologies like electric and hydrogen propulsion. Airbus is a direct competitor across commercial aircraft, defense & space sectors.	- Commercial Aircraft, Defense and space technology, Helicopters - Airbus 72%, Airbus defense and space 17% , airbus helicopters 11%
	EV/Sales: 1,69 P/E: 16,51	EV/Revenue: 1.86 EV/EBITDA: 11.71	Lockheed Martin is a major defense player, recently winning a big F-35 contract and developing hypersonic weapons. Analysts are bullish due to rising defense spending and the F-35 program. Their focus is on core defense and benefits from geopolitical tensions. They are competitor in defense & space sector but not commercial aircraft.	- Aeronautics, Missiles and Fire Control, Rotary and Mission systems, Space - Aeronautics 40%, Rotary and mission systems 26%, space 18%, missile and fire control 16%
	EV/Sales: 1,85 P/E: 15,39	EV/Revenue: 2.17 EV/EBITDA: 11.20	Northrop Grumman is another defense giant, recently awarded the B-21 bomber program. They also make headway in drones. Analysts are positive on their growth potential, particularly in the B-21 program. They focus on core defense and expanding in autonomous tech and space. They are competitors in defense & space sector.	- Aerospace systems, Mission systems, Technology services, Innovation Systems - Space systems 34%, missions' systems 27%, aeronautics systems 26%, defense systems 13%,
 	EV/Sales: 1,62 P/E: 18,81	EV/Revenue: 1.74 EV/EBITDA: 13.59	BAE Systems (merged with Raytheon) is a powerhouse in defense & aerospace. Analysts are optimistic regarding the recent merger and their competitive edge. Their plan is to focus on consolidation in defense and expanding into cyber security. They are competitors in defense & space sector.	- Electronic systems, Cybersecurity, Industrial manufacturing, Maritime - Air 28%, Electronic systems 23%, maritime 23%, platform & services 16%, cybersecurity & intelligence 10%
	EV/Sales: 1,73 P/E: 21,69	EV/Revenue: 1.94 EV/EBITDA: 15.02	General Dynamics balances business jets with its core defense business. They're investing in AI for military applications, with analysts mostly holding due to the solid performance in business jets but some uncertainty in defense. They compete with Boeing in business jets but not large commercial aircraft.	- Aerospace, Combat systems, IT, Marine systems - Technologies 30%, marine systems 29%, aerospace 21%; combat systems 20%

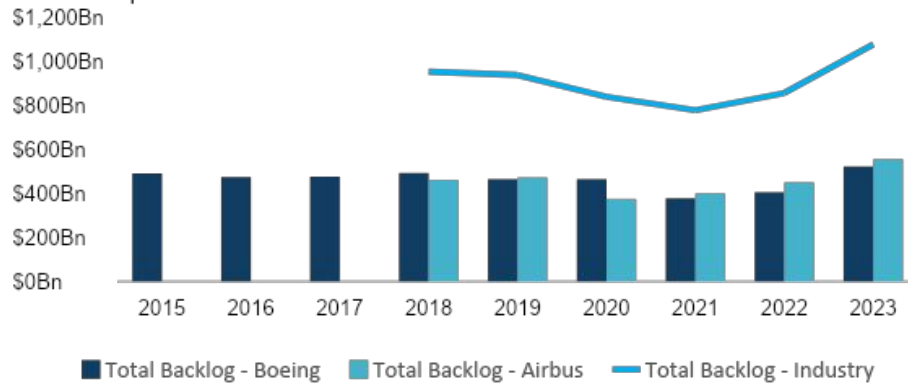


Resilience Amid Challenges: Industry Performance & Recovery

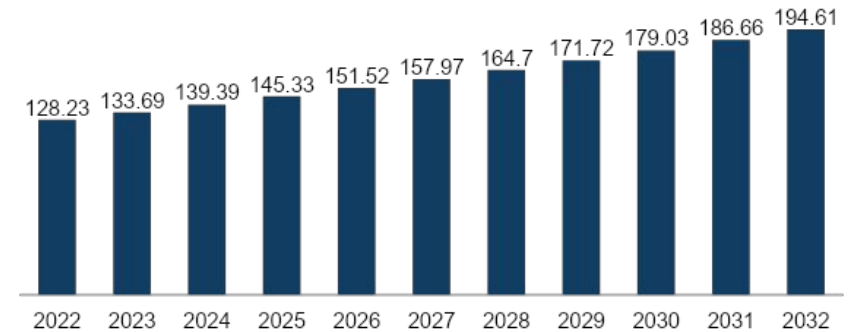
Despite macro-economic challenges, the aviation industry demonstrated resilience, with strong performance driven by pent-up demand. Industry-wide revenue passenger-kilometres (RPKs) reached 87.8% of the 2019 level, and cargo remained a significant revenue contributor.

Key Figures

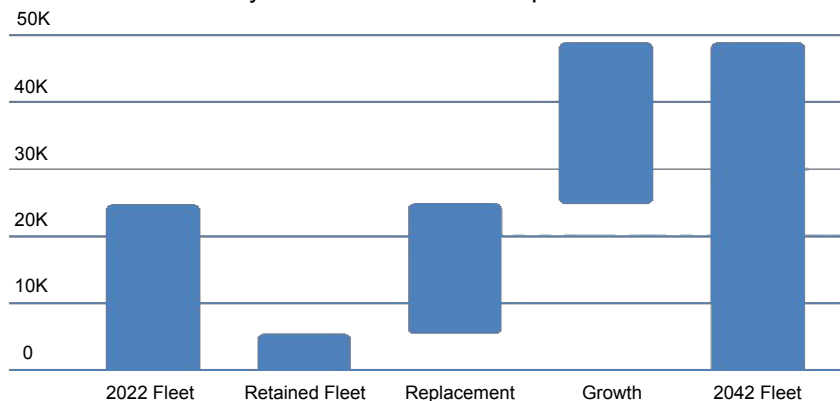
Global backlogs for major aircraft manufacturers has surpassed pre-COVID19 levels



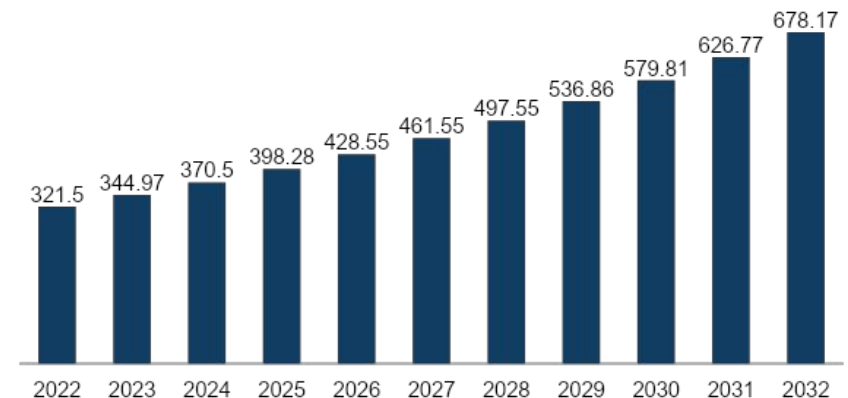
Commercial Aircraft Market Size 2023 - 2032(USD Billion)



Global Outlook 2023-2042: Global fleet will double, nearly half of deliveries for replacement



Aerospace Market Size(USD Billion)





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Investment Thesis

Resilience Amidst Challenges

- Given recent investor panic relating to the grounding of the 737 Max, Boeing Co. (NYSE:BA) is undervalued in the market with potential 24–36-month upside.
- Boeing remains a dominant player in the airline industry and will continue to be so given the high switching costs in the Airline Manufacturing Industry.

Product Improvement and Stability

- Current issues across Boeing's production line are now known and being tackled, future scaling up will be more effective.
- Boeing has temporarily reduced production rates to focus on quality and quality control, putting their production line in better shape.

Short-Term Volatility VS Long-Term Potential

- Even with the airline industry recovering since the pandemic, and the grounding of the 737 Max, Boeing's first quarter performance was relatively unaffected.
- The company has a huge backlog for the next decade, meaning Boeing has incredible upside potential.



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Historical Share Price Analysis

Key Events & Share Price Drivers



- **2011** – Boeing launches the 737 Max program
- **July 2020** – Dennis Muilenburg becomes CEO
- **Oct 2018** – Lion Air crash
- **March 2019** – Ethiopian Airlines crash
- **Jan 2020** – Dave Calhoun becomes CEO
- **Feb 2020**– Covid Crisis
- **Jan 2021**– Boeing settled for \$2.5 Billion in court after fraud allegations
- **Jan 2024** – Grounding of 171 aircrafts after the Alaska Airlines door plug blowout.
- **March 2024** – CEO Dave Calhoun announces he will be stepping down.



Valuation Summary

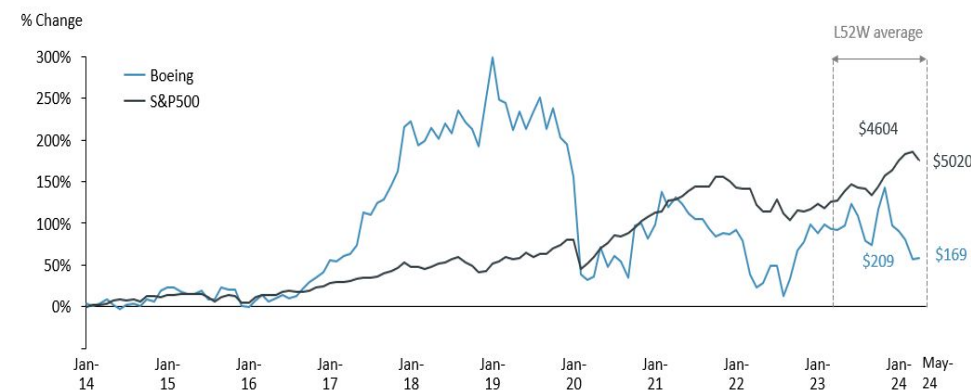
Stock Price Performance

Relative underperformance:

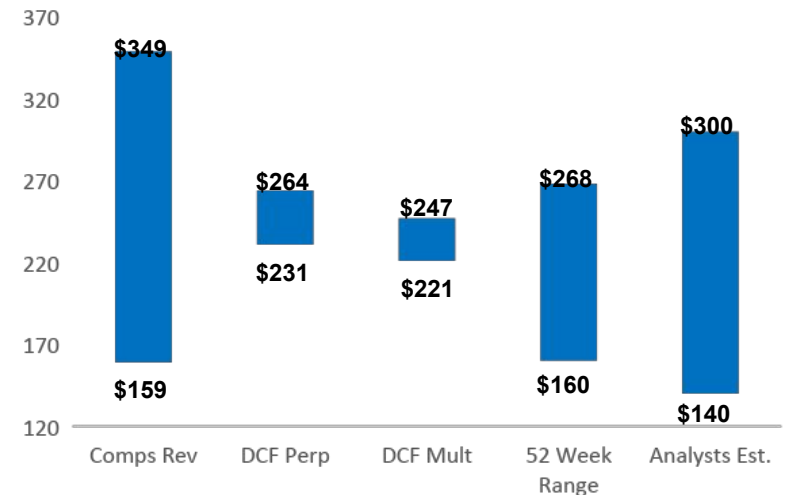
- **YTD:** While the S&P 500 is up around 6% year-to-date, BA stock is down over 32%.
- **3Y:** BA is down roughly 25% compared to a 35% gain for the S&P 500 since 2021.
- **Trading near lows:** BA is currently trading near its 52-week low of \$150, indicating a weak sentiment.

Market sentiment:

- **Priced-in catalysts:** The current price reflects Boeing's issues faced with the 737 MAX and expectations for increased commercial airplane deliveries in the coming years.
- **Loss of confidence:** The stock price is not reflecting the strong Q1 revenue that beat analyst estimates. This could be due to loss in confidence and the wait for a full recovery in commercial airplane production.



Valuation Football Field



Broker Estimates	Target Prices (\$)	Upside (%)	Rating
Citigroup (25/04/2024)	\$224	+32.55%	Strong Buy
Deutsche Bank (25/04/2024)	\$225	+33.14%	Strong Buy
Barclays (22/04/2024)	\$190	+12.43%	Hold
JP Morgan (17/04/2024)	\$230	+35.78%	Strong Buy
RBC Capital (16/04/2024)	\$215	+27.23%	Buy



Comparables Analysis

Comparable Company Rationale

These companies operate in the aerospace and defence sector, making them direct competitors or collaborators with Boeing.

- **Key Similarities and Differences:** All focus on aerospace/defence manufacturing and rely on government contracts. Companies like General Electric have more diversified portfolios, while TransDigm specializes in aerospace components, contrasting with Boeing's comprehensive aircraft manufacturing.
- **Boeing** can enhance safety measures and diversify technology to regain market trust and strengthen competitiveness against industry leaders like Lockheed.

(in millions of U.S. dollars)		Stock	Equity	Firm	TEV / Revenue				TEV / EBITDA				Price / Earnings			
Company	Symbol	Price	Value	Value	2021	LTM	NTM	2025	2021	LTM	NTM	2025	2021	LTM	NTM	2025
Northrop Grumman	NYSE:NOC	485.03	\$71,977	\$86,727	1.9x	2.2x	2.09x	2.11x	12.11x	18.70x	18.95x	15.38x	8.59x	14.35x	19.39x	19.64x
RTX Corporation	NYSE:RTX	101.52	\$134,353	\$174,652	2.4x	2.5x	2.18x	2.21x	18.30x	17.80x	14.35x	13.69x	31.8x	37.48x	18.45x	18.77x
General Electric	NYSE:GE	161.82	\$174,808	\$178,205	2.0x	2.6x	4.74x	4.9x	36.20x	18.40x	16.00x	22.51x	12.9x	19.38x	37.73x	39.53x
L3Harris Technologies, Inc.	NYSE:LHX	214.05	\$40,164	\$53,459	2.7x	2.8x	2.49x	2.52x	13.10x	14.70x	15.50x	13.93x	23.25x	17.32x	15.99x	16.33x
General Dynamics	NYSE:GD	287.09	\$78,487	\$88,514	1.8x	2.1x	1.85x	1.89x	13.30x	17.00x	16.70x	14.78x	17.07x	23.88x	18.74x	19.64x
Lockheed Martin	NYSE:LMT	464.93	\$110,786	\$127,414	1.6x	1.8x	1.81x	1.83x	12.20x	12.20x	12.30x	13.03x	14.71x	16.88x	17.42x	17.62x
TransDigm Group	NYSE:TDG	1,248.03	\$69,470	\$87,208	10.7x	12.5x	10.98x	11.30x	27.30x	25.40x	21.10x	21.67x	46.60x	42.92x	37.96x	39.22x
Textron Inc.	NYSE:TXT	84.59	\$16,165	\$18,685	1.5x	1.4x	1.27x	1.29x	13.80x	10.60x	11.40x	10.62x	23.26x	18.11x	12.85x	13.46x
Airbus SE	ENXTPA:AIR	165.2	\$130,663	\$127,906	1.4x	1.8x	1.62x	1.68x	21.00x	20.60x	24.40x	18.80x	21.11x	27.50x	22.66x	24.42x
Boeing Company	NYSE:BA	\$167.84	\$103,034	\$143,438	2.6x	1.9x	1.8x	2.6x	10.5	46.50x	23.10x	29.20x	NM	NM	NM	NM
High			\$134,353	\$180,526	10.7x	12.5x	10.9x	11.3x	36.2x	25.4x	24.4x	21.7x	46.6x	42.9x	37.9x	39.5x
Median			\$54,999	\$70,212	2.4x	2.3x	1.9x	2.2x	13.5x	18.1x	16.35x	15.0x	101.5	21.1x	18.7x	19.6x
Mean			\$62,219	\$71,420	2.8x	2.9x	2.9x	3.2x	16.7x	20.2x	17.4x	17.4x	103.8	22.9x	22.4x	23.1x
Low			\$16.165	\$18,685	1.4x	1.4x	1.2x	1.29x	12.1x	10.6x	11.40x	13.0x	12.9x	17.3x	12.8x	13.5x



DCF Model Output

Valuation Summary

Terminal Value - Multiples Method

Implied Share Price from DCF	\$234.34	Premium to Current Share Price	39.6%
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Median TEV / EBITDA of Comps	13.5x	Implied Enterprise Value	\$199,322
Baseline Terminal EBITDA Multiple	13.0x	% of Implied TEV from TV	51.8%
Baseline Terminal Value	\$256,817	Total Debt & Capital Leases	(47,940)
PV of Terminal Value	\$103,157	Implied Equity	\$143,860
Sum of PV of FCFs	\$96,165	Diluted Shares Outstanding	613.9M

Terminal Value – Perpetuity Growth Method

Implied Share Price from DCF	\$240.80	Premium to Current Share Price	43.5%
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Expected Long-Term GDP Growth	1.50%	Sum of PV of FCFs	\$96,165
Baseline Terminal FCF Growth Rate	0.50%	Implied Enterprise Value	\$203,280
Baseline Terminal Value	\$266,671	% of Implied TEV from TV	52.7%
Implied Terminal EBITDA Multiple	13.5x	Total Debt & Capital Leases	(47,940)
PV of Terminal Value	\$266,671	Implied Equity Value	\$147,818

WACC Assumptions

WACC
9.55%

Cost of Equity
11.80%

Cost of Debt
5.61%

Risk Free Rate
4.69%

Equity Risk Premia
4.60%

Sensitivity Analysis

Terminal EBITDA Multiple:

WACC		11.4x	12.2x	13.0x	13.8x	14.6x
	10.05%	\$ 203	\$ 213	\$ 223	\$ 233	\$ 242
	9.80%	\$ 208	\$ 218	\$ 228	\$ 239	\$ 249
	9.55%	\$ 214	\$ 224	\$ 234	\$ 245	\$ 255
	9.30%	\$ 219	\$ 230	\$ 240	\$ 251	\$ 262
	9.05%	\$ 225	\$ 236	\$ 246	\$ 257	\$ 268

Terminal FCF Growth Rate:

WACC		0.00%	0.25%	0.50%	0.75%	1.00%
	10.05%	\$ 211	\$ 215	\$ 220	\$ 224	\$ 229
	9.80%	\$ 220	\$ 225	\$ 230	\$ 235	\$ 240
	9.55%	\$ 231	\$ 236	\$ 241	\$ 246	\$ 252
	9.30%	\$ 241	\$ 247	\$ 252	\$ 258	\$ 264
	9.05%	\$ 252	\$ 258	\$ 264	\$ 270	\$ 277



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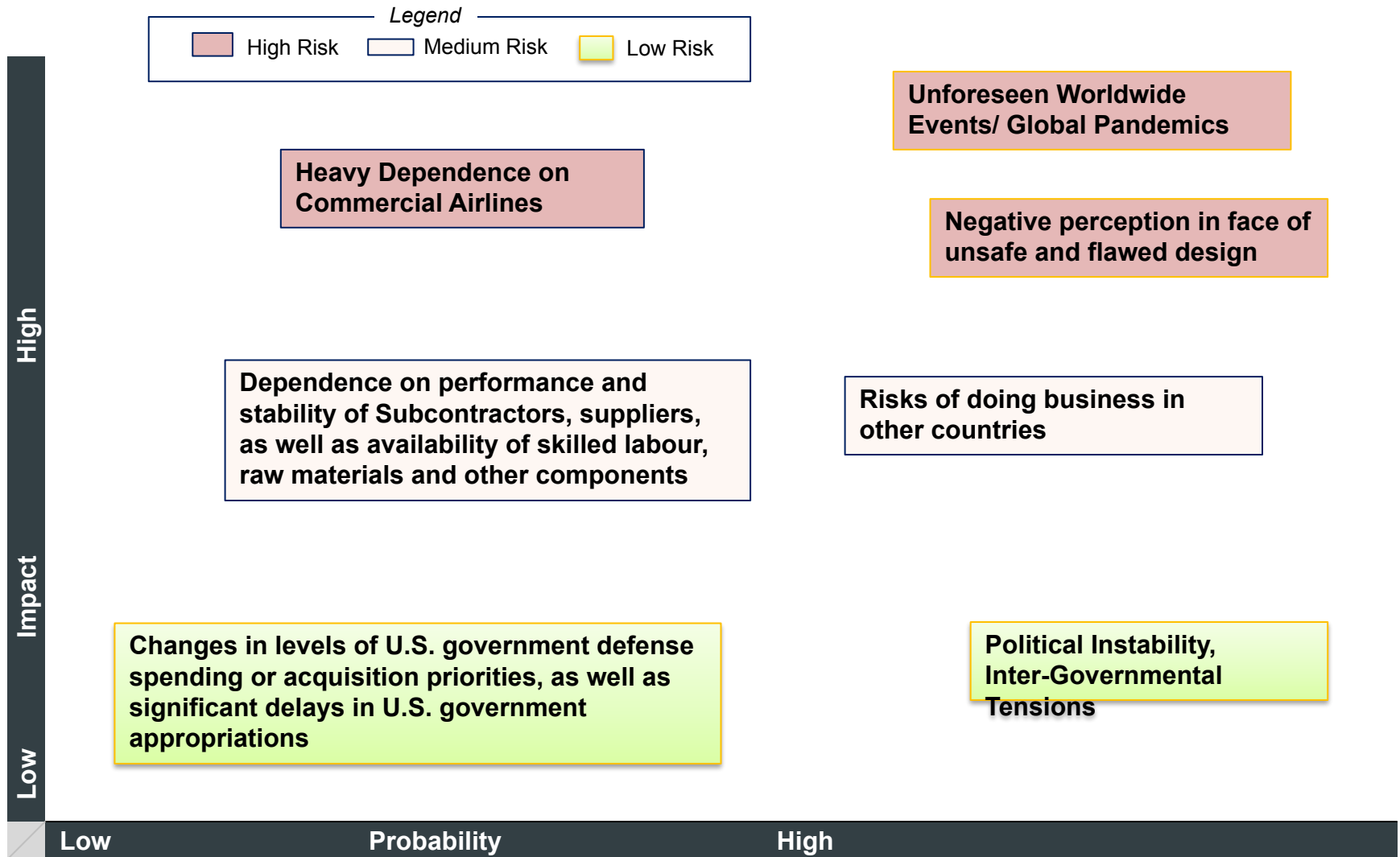
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Risk Considerations





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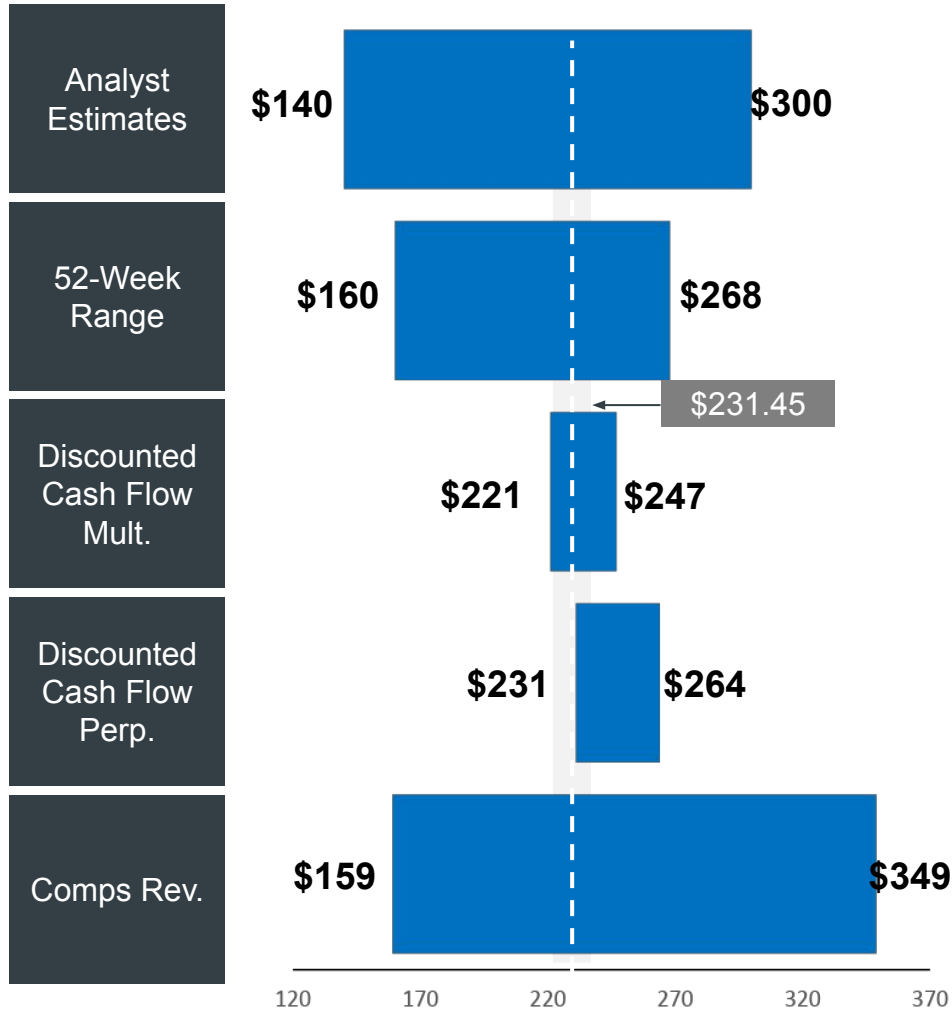
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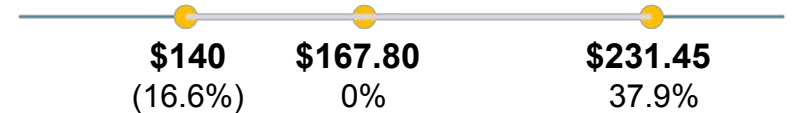
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Price Target & Recommendation



Price Target		
Valuation Summary		
Analysis	Price	Weight
Comparables	221.00	30%
Perpetuity DCF	240.80	30%
Multiple DCF	234.30	30%
Street	226.10	10%
Target Price	231.45	100%



The stock should be bought at below \$178